



Final
MISO DPP 2018 April South Area Study
Phase III Report

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1. Executive Summary

This report presents the results of a System Impact Study performed to evaluate the interconnection of the generators in the DPP 2018 April South Area Phase III (South Area DPP 3). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Entergy, Cooperative Energy, CLECO, AECL, MLGW, PowerSouth, Southern Company (SOCO), SPP, and TVA.

1.1. Project List

The original interconnection requests for DPP 2018 April South Area had a total of 49 generation projects. Projects J1029, J1133, J1134, J1136, and J1167 withdrew prior to the start of DPP Phase I. Projects J961, J964, J972, J988, J1023, J1049, J1070, J1073, J1078, J1079, J1091, J1093, J1095, J1097, J1099, J1100, J1113, J1120, J1130, J1137, J1151, J1156, J1157, J1159, J1162, J1163, J1165, J1166, J1168, and J1186 withdrew during Decision Point 1. During Decision Point 1 Project J962 reduced their NRIS request from 100 MW to 0 MW and Project J1147 reduced their NRIS request from 100 MW to 0 MW. J1020 and J1147 withdrew during Decision Point 2. During Decision Point 2 Project J1007 reduced their ERIS and NRIS requests from 200 MW to 180 MW, and Project J1060 reduced their ERIS from 500 MW to 450 MW and their NRIS request from 500 MW to 400 MW. Therefore, there are 12 generation projects with a combined nameplate rating of 1,611 MW of Energy Resource Interconnection Service (ERIS) & 1,311 MW of Network Resource Interconnection Service (NRIS). The detailed list of South Area DPP 3 is shown below in Table 1, and the Phase III study was kicked off on January 27th, 2020.

Table 1: List of DPP 2018 April South Area Phase III Projects

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	Point of Interconnection (POI)
J1007	Solar	EES	Crittenden	AR	NRIS	180	Marion - Marked Tree 161 kV Line
J1030	Solar	EES	Liberty	TX	ERIS	150	Gulf Livingston - Menard 138 kV Line
J1060	Solar	EES	St. Francis	AR	NRIS	450/400*	West Memphis - Keo 500 kV Line
J1076	Solar	EES	St. James Parish	LA	NRIS	200	Vacherie 230 kV Substation
J1125	Solar	EES	Cross	AR	NRIS	135	Wynne South - AECC Wynne E 161 kV Line
J1142	Solar	EES	St. James Parish	LA	NRIS	20	Vacherie 230 kV Substation
J1144	Solar	CE	Covington/Lamar	MS	NRIS	96	Moselle - Columbia 161 kV Line
J1146	Coal	EES	Mississippi	AR	NRIS Only	10	Sans Souci 500 kV Substation
J1155	Solar	EES	Mississippi	AR	NRIS	150	Driver 230 kV Substation
J1158	Solar	EES	St. James Parish	LA	NRIS	70	Vacherie 230 kV Substation
J1184	Solar	EES	Washington Parish	LA	NRIS	50	Lagen Holton - Franklinton 115 kV Line
J962	Solar	EES	DeSoto	MS	ERIS	100	Twinkletown 230 kV Substation

*450MW of ERIS, 400MW of NRIS

1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2018 April South Phase III reflects responsibilities for mitigating system impacts. The total cost of Network Upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in Appendix A – Cost Allocation Summary (CEII).

Table 2: Total Cost of Network Upgrades for DPP 2018 April South Phase III Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)	M4 (\$)
	Steady State	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned					
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i	\$4,000/MW	(10% of (j) from Phase I) – M2	(20% of (j) from Phase II) – M2 – M3
J1007	0	0	0	1,456	0	15,241,000	1,552,000	357,242	15,598,242	800,000	1,849,571	5,675,213
J1030	0	0	0	0	0	16,085,000	1,079,000	0	16,085,000	600,000	1,088,306	1,111,694
J1060	0	0	0	300,000	0	35,192,000	2,852,000	0	35,192,000	2,000,000	1,395,904	15,967,049
J1076	0	0	0	0	1,724,138	228,000	1,148,000	0	1,952,138	800,000	0	1,288,009
J1125	0	0	0	2,494	0	14,316,000	1,448,000	564,476	14,880,476	540,000	3,032,486	2,573,987
J1142	0	0	0	0	172,414	228,000*	1,101,500*	0	400,414	80,000	0	0
J1144	0	0	0	0	0	0	5,251,585	0	0	384,000	348,156	0
J1146	0	0	0	0	0	275,000	0	0	275,000	40,000	2,586	30,596
J1155	0	0	0	0	0	283,000	2,666,000	0	283,000	600,000	662,449	0
J1158	0	0	0	0	603,448	228,000	1,101,500	0	831,448	280,000	414,402	843,712
J1184	0	0	0	0	0	2,377,000	91,000	0	2,377,000	200,000	0	0
J962	0	0	0	0	0	440,000	2,377,000	0	440,000	400,000	1,364,893	441,925

*These figures represent expected costs and differ from those that appear in J1142's PGIA. PGIA costs are valid until GIA execution.

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. ERIS Network Upgrades and NRIS Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.

Table 3: ERIS & NRIS Upgrades (Planning-level cost estimates)

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Tezcuco - Frisco 230 kV #1 & #2 Re-conductor	EES		J1076, J1142, J1158	2,500,000

Table 4: Shared Network Upgrades (Planning-level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
Kuhn Road – Ebony 161 kV Rebuild	EES	J934	J1007, J1125	4,161,000

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) Analysis performed shows there are two projects for Shared Network Upgrade cost allocation.
- 4) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, states that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all wind and solar projects included in the DPP 2018 April South study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter-based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading and lagging power factor, as shown below in Table 5.

Table 5: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f. (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1007	200	±65.737	2 x 14 MVar Cap	197.6	93.2	0.9044	196.8	110.6	-0.8718	Yes	
J1030	153.5	±50.453	2 x 16 MVar Cap	150.6	56.1	0.9371	147.3	124.0	-0.7650	Yes	
J1060	500	±242.161	2 x 6 MVar Cap	488.9	210.0	0.9188	485.8	287.5	-0.8606	Yes	
J1076	200	±96.864	N/A	197.3	66.7	0.9473	195.8	147.7	-0.7983	Yes	
J1125	135	±101.25	N/A	133.1	73.0	0.8768	130.3	171.3	-0.6054	Yes	
J1142	20	±11.222	2 x 4.8 MVar Cap	19.6	17.7	0.7422	19.4	18.7	-0.7200	Yes	
J1144	96	±72	1 x 5 MVar Cap	91.9	61.4	0.8315	88.4	102.3	-0.6538	Yes	
J1155	150	±72.6	2 x 24 MVar Cap	147.6	102.0	0.8227	147.1	87.1	-0.8605	Yes	
J1158	70	±33.9	2 x 11 MVar Cap	68.9	43.8	0.8439	68.2	56.7	-0.7690	Yes	
J1184	50	±24.527	2 x 7 MVar Cap	49.3	30.3	0.8520	48.9	41.5	-0.7624	Yes	
J962	100	±32.868	2 x 13.5 MVar Cap	98.9	45.1	0.9099	98.6	38.5	-0.9315	Yes	

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2018 April South models is the MTEP18 models with the Bench Cases including all pre-queued projects and associated Network Upgrades, while the Study Cases contain all of the interconnection requests in DPP 2018 April South Phase III, in addition to all the facilities in the Bench Cases.

- Bench Cases
 - APR18-PhaseIII-2023SUM-Bench
- Study Cases
 - APR18-PhaseIII-2023SUM-Study

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency



rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2018 April South generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.11.0 and TARA version 1901.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has a distribution factor (DFAX) greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting NRIS must mitigate constraints found by using the Deliverability algorithm to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2023 Summer show generator project J1144 causing constraints. The details pertaining to the thermal analysis can be found in Appendix C – MISO ERIIS Analysis (CEII).



4.1. J1144

Thermal analysis determined that this generator contributes to the following constraints:

1. J1144 POI – Columbia 161 kV

The Moselle – Columbia 161 kV line tapped by project J1144 is de-rated due to clearance violations at Moselle. However, per Cooperative Energy’s guidance, the J1144 – Columbia segment of this line will be operated at a higher rating after the interconnection of J1144, and no additional network upgrade is required.

5. Voltage Analysis

The voltage analysis results for 2023 Summer show two voltage constraints that are mitigated by MTEP Project 15729, Appendix A in MTEP19. No additional mitigations are required and no study project is assigned any cost for the mitigation of said constraints. The details pertaining to the voltage analysis can be found in Appendix C – MISO ERIS Analysis (CEII).

6. Stability Analysis

Stability analysis shows that the study projects did not adversely impact the system. An additional stability study capturing the Entergy Local Planning Criteria (LPC) for new generation interconnections was also performed by MISO and shows that study projects did not adversely impact the system under the Entergy LPC for new generation interconnections. The details pertaining to the stability analysis can be found in Appendix D – MISO Stability Analysis (CEII) and Appendix E – Entergy Local Planning Criteria Stability Analysis (CEII).

6.1. Model Development

The following summer peak 2023 models were developed based on Phase III study models. The Entergy LPC stability models were developed based on the Phase III stability study models and were adjusted in order to comply with Entergy’s LPC by fully dispatching nearby local generation.

- Bench Case
 - APR18-PhaseIII-2023SUM-Bench
 - APR18-PhaseIII-2023SUM-Bench_Entergy LPC
- Study Case
 - APR18-PhaseIII-2023SUM-Study
 - APR18-PhaseIII-2023SUM-Study_Entergy LPC

6.2. Study Methodology

The purpose of the study is to identify potential angular instabilities, voltage dip violations, and damping violations, if any, due to the interconnection of the projects in the DPP 2018 April South study cycle under disturbance conditions, and the impact of all study projects on the system stability performance.



The fault scenarios simulated in this study cover faults simulated as part of the MTEP18 analysis as well as selected three-phase (3PH) faults with normal clearing and single line to ground (SLG) faults with delayed clearing. Dynamic simulations of fault scenarios were performed using the DSATools TSAT program (version 18.0.10).

Fault scenarios were first simulated using the study case and the results were reviewed. For scenarios that exhibited instability, the bench case was simulated such that the stability performance with and without the proposed interconnection projects could be compared. Any new stability problems attributed to the proposed interconnection projects are flagged and reported.

For each fault, rotor angles, speed deviation, and electrical power outputs of the study generators and the generators in the proximity were monitored. Voltages at selected buses, including all POI buses of the study projects and all future buses, were also monitored.

In addition Section 10 of the “Entergy Transmission Local Planning Criteria”¹ states the following:

- **New Generator Interconnections:** *For Generator Interconnection (GI) Stability Studies and TPL Reliability Stability Studies only, new generation requests (for GI studies) or generation with an executed Generator Interconnection Agreement (GIA) since performance of the last reliability assessment (for TPL studies) will be studied with all generation local to the new generator at full output to determine whether the aggregate of the generation can be injected into the transmission system without experiencing transient stability issues for local faults.*

For the purposes of the Entergy LPC study local generation has been defined as generation located within one breaker substation from the Point of Interconnection if interconnecting to an existing substation or located within one breaker substation from a termination substation when tapping a line. A summary of the generation dispatch for the Entergy LPC analysis has been tabulated in Appendix E – Entergy Local Planning Criteria Stability Analysis (CEII).

The fault scenarios simulated in the Entergy LPC study cover selected 3PH and 3PH P3 contingency faults with normal clearing. In each fault scenario, a generator is disconnected at 0.5 seconds and the simulation continues to run until 5.0 seconds at which point the fault is initiated and the total simulation run time is 20.0 seconds.

6.3. Study Criteria

Transient stability study criteria are based upon Section 11.A of the “Entergy Transmission Local Planning Criteria”.

In addition, all renewable study projects are subject to the voltage ride-through and frequency ride-through criteria specified in NERC PRC-024-2 (“Generator Frequency and Voltage Protective Relay Settings”²) to check if the projects remain connected during frequency and voltage excursions. Specifically, PRC-024-2 mandates that protective relaying should be set in such a way that:

- Voltage Ride-Through: a generator shall withstand zero voltage at the POI (typically the primary side of the station transformer) for up to 0.15 seconds (9 cycles) and the ensuing voltage recovery period for three phase faults.
- Frequency Ride-Through: a generator shall maintain continuous operation between 59.5 and 60.5 Hz.

¹ <https://cdn.misoenergy.org/Entergy%20TO%20Planning%20Criteria108226.pdf>

² <http://www.nerc.com/pa/Stand/Reliability%20Standards/PRC-024-2.pdf>

6.4. Study Results

The stability analyses showed that the study projects did not adversely impact the system. The complete list of 3PH and SLG faults simulated as well as their corresponding results and plots are included in Appendix D – MISO Stability Analysis (CEII) and Appendix E – Entergy Local Planning Criteria Stability Analysis (CEII).³

7. Short Circuit Analysis

Short circuit analysis was performed utilizing ASPEN software. Single line to ground faults and three phase faults were evaluated for pre- and post-project cases (similar to Bench and Study models). The short circuit studies for the DPP 2018 April South study cycle indicate that the study generators do not have adverse impacts on circuit breaker capability. The details pertaining to the short circuit analysis are presented in Appendix F – Short Circuit Study Reports (CEII).

8. Affected System Impact Study

The details pertaining to the AECI, MLGW, PowerSouth, SOCO, TVA, and SPP Affected Systems studies are in Appendix G – AECI Affected Systems Study Report (CEII), Appendix H – MLGW Affected Systems Study Report (CEII), Appendix I – PowerSouth Affected Systems Study Report (CEII), Appendix J – SOCO Affected Systems Study Report (CEII), Appendix K – TVA Affected Systems Study Report (CEII), and Appendix L – SPP Affected Systems Report (CEII). PowerSouth, SOCO, TVA, and SPP did not identify any constraints for mitigation for projects in Phase III.

8.1. J1007

The AECI Study identified that this generator contributes to the following constraint:

1. Palmyra 345/161 kV transformer

Per AECI cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add a second 345/161 kV transformer at Palmyra with a cost estimate of \$9.3 million. The project is allocated 0.016% of the cost.

8.2. J1030

No affected systems mitigations were found to be required for this generator.

8.3. J1060

The MLGW Study identified that this generator contributes to the following constraint:

1. Shady Grove – N.E. Gate 161 kV

Per MLGW cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to uprate the Shady Grove – N.E. Gate 161 kV line with a cost estimate of \$300,000. The project is allocated 100% of the cost.

8.4. J1076

No affected systems mitigations were found to be required for this generator.

8.5. J1125

The AECI Study identified that this generator contributes to the following constraint:

³ J1184 is preparing a Material Modification for its inverters (Power Electronics -> TMEIC) during Phase III of the DPP-APR-18 cycle. MISO obtained the dynamics data for the new inverters (TMEIC) and used that data in the simulation of the 5087_S_EES_P7_SLG fault and showed that excess reactive power and high voltage at the generator bus issues were mitigated. A test of the full suite of faults will be obtained through J1184's Material Modification study and issues identified therein may need remediation.



1. Palmyra 345/161 kV transformer

Per AECI cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add a second 345/161 kV transformer at Palmyra with a cost estimate of \$9.3 million. The project is allocated 0.027% of the cost.

8.6. J1142

No affected systems mitigations were found to be required for this generator.

8.7. J1144

No affected systems mitigations were found to be required for this generator.

8.8. J1146

No affected systems mitigations were found to be required for this generator.

8.9. J1155

No affected systems mitigations were found to be required for this generator.

8.10. J1158

No affected systems mitigations were found to be required for this generator.

8.11. J1184

No affected systems mitigations were found to be required for this generator.

8.12. J962

No affected systems mitigations were found to be required for this generator.

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (monitored facilities – contingency facility pairs) seen in the deliverability analysis for



the 2023 Summer case are summarized in Appendix M – MISO Deliverability Analysis (CEII).

9.3.1. J1007

This generator is found to be fully deliverable up to 180 MW without any Network Upgrades.

9.3.2. J1060

This generator is found to be fully deliverable up to 400 MW without any Network Upgrades.

9.3.3. J1076

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 6 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J1076

J1076 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuco – Frisco 230 kV #1	0.0	12.5%	J1076, J1142, J1158	863,797	2,500,000
Tezcuco – Frisco 230 kV #2	200.0	12.45%	J1076, J1142, J1158	860,341	

9.3.4. J1125

This generator is determined to be fully deliverable up to 135 MW without any Network Upgrades.

9.3.5. J1142

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J1142

J1142 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuco – Frisco 230 kV #1	0.0	12.5%	J1076, J1142, J1158	86,380	2,500,000
Tezcuco – Frisco 230 kV #2	20.0	12.45%	J1076, J1142, J1158	86,034	

9.3.6. J1144

This generator is found to be fully deliverable up to 96 MW without any Network Upgrades

9.3.7. J1146

This generator is found to be fully deliverable up to 10 MW without any Network Upgrades

9.3.8. J1155

This generator is found to be fully deliverable up to 150 MW without any Network Upgrades.

9.3.9. J1158

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J1158

J1158 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuco – Frisco 230 kV #1	0.0	12.5%	J1076, J1142, J1158	302,329	2,500,000
Tezcuco – Frisco 230 kV #2	70.0	12.45%	J1076, J1142, J1158	301,119	

9.3.10. J1184

This generator is found to be fully deliverable up to 50 MW without any Network Upgrades.

10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis, which tests for Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study.

The maximum MW impacts and SNU cost allocations appear in Table 9.

Table 9: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
Kuhn Rd – Ebony 161 kV Rebuild	DPP-2017-AUG	J934	133.02	4,161,000	3,239,283
	DPP-2018-APR	J1007	14.67		357,242
	DPP-2018-APR	J1125	23.18		564,476

11. Cost Allocation

The cost allocation of Network Upgrades (NU) for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of NRIS as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each NU identified in the System Impact Study is provided by the corresponding Transmission Owner.

11.2. Cost Allocation Methodology

The costs of NU for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.



Cost Allocation Methodology for Thermal Constraints

1. With all study group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any Network Upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. The cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Project A Cost Portion of NU} = \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D – MISO Stability Analysis (CEII)

Appendix E – Entergy Local Planning Criteria Stability Analysis (CEII)

Appendix F – Short Circuit Study Reports (CEII)

Appendix G – AECI Affected Systems Study Report (CEII)

Appendix H – MLGW Affected Systems Study Report (CEII)

Appendix I – PowerSouth Affected Systems Study Report (CEII)

Appendix J – SOCO Affected Systems Study Report (CEII)

Appendix K – TVA Affected Systems Study Report (CEII)

Appendix L – SPP Affected Systems Study Report (CEII)

Appendix M – MISO Deliverability Analysis (CEII)